



A note on using random starting values in small sample SEM

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Abstract

Model estimation for SEM analyses in commonly used software typically involves iterative optimization procedures, which can lead to nonconvergence issues. In this paper, we propose using random starting values as an alternative to the current default strategies. By drawing from uniform distributions within data-driven lower and upper bounds (see De Jonckere et al. (2022) *Structural Equation Modeling: A Multidisciplinary Journal*, 29(3), 412–427), random starting values are generated for each (free) parameter in the model. Through three small simulation studies, we demonstrate that incorporating such bounded random starting values significantly reduces the nonconvergence rate, resulting in increased convergence rates ranging between 87% and 96% in the first two studies. In essence, bounded random starting values seem to offer a promising alternative to the default starting values that are currently used in most software packages.

Keywords SEM · Small sample size · Starting values · Convergence

Introduction

Structural equation modeling (SEM) is a statistical method that allow testing (complex) relationships among latent variables. A prominent challenge in SEM is nonconvergence, which occurs when the optimizer fails to reach a stable solution. This issue is particularly prevalent when dealing with small sample sizes. Over the years, several solutions that try to tackle the problem of nonconvergence have been proposed. For instance, recent developments show that imposing simple bounds on the model parameters during estimation drastically decreases the amount of nonconverged cases and this without negatively affecting the point estimates of the unbounded parameters (De Jonckere and Rosseel, 2022; Lüdtke et al., 2021). The simple lower and upper bounds proposed by the authors are (by default) derived directly from the data and are typically calculated for the (residual) (co)variances of the observed variables, the (residual) (co)variances of the latent variables, and the factor loadings.

An alternative to address the issue of nonconvergence is to use Bayesian estimation. Within the Bayesian framework, a crucial step involves specifying prior distributions for the model parameters. However, the process of selecting appropriate priors is often intricate. In fact, using default or diffuse priors can sometimes lead to worse performance compared to standard maximum likelihood estimation (Smid and Rosseel, 2020; Smid, 2023). Therefore, it is often recommended to perform a sensitivity analysis to assess the impact of those priors on the results (Van Erp et al., 2018).

Somewhat related to Bayesian estimation is shrinkage estimation of the sample variance-covariance matrix. In shrinkage estimation, a weighted average of the sample variance-covariance matrix and a highly structured shrinkage target is calculated. De Jonckere and Rosseel (2023) constructed a rationale for a model-based shrinkage target specifically designed for SEM analyses. This approach also allows researchers to incorporate their prior knowledge about the data, which can be translated directly into the model-based target matrix (De Jonckere and Rosseel, 2023). As with Bayesian estimation, the accuracy of this prior knowledge is crucial for the performance of the model-based shrinkage target. In their study, De Jonckere and Rosseel (2023) demonstrated that the use of the model-based shrinkage target can effectively reduce the number of nonconverged cases compared to standard maximum likelihood estimation.

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Nonconvergence cannot be solely attributed to a small sample size, as datasets with equivalent sample sizes may yield divergent results, with one dataset converging while another fails to do so. A crucial aspect of iterative optimization algorithms for model estimation is the determination of suitable (or optimal) starting values for the model parameters. Various strategies can be employed to determine these starting values. One approach involves adopting a “simple” approach whereby constants are selected as starting values. For example, starting values for factor loadings and (residual) variances can be set to 1, while all other parameters can be set to 0. Alternatively, other ad hoc values can be assigned such as assigning a value of 0.05 for latent variances or half the observed variance for residual variances of the indicators. Another approach involves utilizing noniterative estimators to estimate (a subset of) the model parameters, and use these estimates as starting values. For the measurement part of the model, various noniterative estimation procedures exist such as the Multiple Group Method (Guttman, 1952), Häggglund’s instrumental variable approaches, i.e., FABIN1, FABIN2, and FABIN3 (Häggglund, 1982),¹ the noniterative estimation procedure of Bentler (1982), among others (see Dhaene and Dhaene (2023) for an evaluation of noniterative estimators in CFA). A final noteworthy approach is the use of *random* starting values, where values are somehow randomly selected for each free parameter in the model. The idea of using random starting values is a common one, and this strategy is used in many contexts. For example, the method is often applied in mixture (SEM) models to mitigate the risk of converging to a local maximum or minimum (Shireman et al., 2015; Biernacki et al., 2003). On the other hand, Gilli and Schumann (2010) investigated the best-of-N-restarts strategy with random starting values in the field of finance and economics. A comparison between the best-of-N-restarts strategy (where N was set to 100) and a more “thought through” approach using constants that reflect reasonable values for the financial interpretation of the parameters in the model (see Manousopoulos and Michalopoulos, 2009, p 598), revealed that the utilization of random starting points yielded outcomes that were at least equivalent to, if not superior to, the strategy using constants.

Despite its prevalence in various contexts, random starting values are seldom employed in regular (non-mixture) SEMs. While generating random starting values may appear straightforward, generating ‘good’ random starting values can be challenging. Careful attention must be given to the selection of these values, including the choice of distribution from which they are drawn. Additionally, random starting

values should be chosen to preserve the positive definiteness of the implied covariance matrix of the model. In this study, we propose choosing random starting values from a uniform distribution with a lower and upper bound. Defining the lower and upper bounds for the uniform distributions will be based on the work of De Jonckere and Rosseel (2022). Where, previously, these data-driven bounds were imposed on the model parameters *during* the estimation phase, in the current paper, we propose calculating these data-driven bounds and use them as boundaries for the uniform distributions from which random starting values are drawn, *pre* estimation.

To the best of our knowledge, a review of the existing literature did not yield any papers that specifically investigate the process of defining starting values for regular (non-mixture) SEMs. Therefore, the primary objective of this paper is to examine the impact of selecting different sets of random starting values on the convergence status of a model. It should be stressed that the sole purpose of using these random starting values is to achieve convergence. However, there is no guarantee that this will result in an admissible solution. At the very least, researchers now have a solution to evaluate. The rest of the paper is organized as follows: subsequent sections of this paper will explain the concepts of model estimation in structural equation modeling and optimization algorithms. Furthermore, we will introduce the application of bounds to define initial parameter estimates and explore their effect on the convergence status of different models in three small simulation studies.

Model estimation in structural equation modeling

Assuming multivariate normal and complete data, $\mathcal{Y} \sim \mathcal{N}(\mu, \Sigma)$, the sample variance-covariance matrix, $\mathbf{S}$, and the mean vector $\bar{\mathbf{y}}$ are sufficient statistics. Because in this manuscript we assume centered data (i.e., $\mu = \bar{\mathbf{y}} = \mathbf{0}$), $\mathbf{S}$ is the only summary statistic we need. A discrepancy function $F(\mathbf{S}, \Sigma(\theta))$ expresses the difference between the sample variance-covariance matrix and the model-implied variance-covariance matrix $\Sigma(\theta)$, where θ denotes the free parameters in the model. Model estimation aims at finding those values for θ that minimize the distance between $\Sigma(\theta)$ and $\mathbf{S}$. Different discrepancy functions based on the least squares principle exist, but the maximum likelihood discrepancy function is probably the most commonly used method in SEM. This discrepancy function can be written as:

$$F_{ML}(\theta) = \log|\Sigma(\theta)| - \log|\mathbf{S}| + \text{tr}(\mathbf{S}\Sigma(\theta)^{-1}) - P, \quad (1)$$

where $|\cdot|$ denotes the determinant of a matrix, $\text{tr}(\cdot)$ the trace of a matrix (i.e., sum of its diagonal elements), and P the number of observed variables. Model estimation in SEM as

¹ In lavaan 0.6-15, a combination of the “simple” approach and the approach using noniterative estimators is applied per default. The factor loadings are initially estimated using FABIN3 (Häggglund, 1982), the observed residual variances are set to half the observed variance ($\frac{1}{2} \text{Var}(y_p)$) and the latent variances are set to 0.05. Covariances and regression coefficients are initialized to zero.

Table 1 Bounds for UVI

Parameter	Lower bound	Upper bound
$\text{Var}(f)$	1	1
λ_p	$-\sqrt{\text{Var}(y_p)}$	$\sqrt{\text{Var}(y_p)}$
$\text{Var}(\epsilon_p)$	0	$\text{Var}(y_p)$

Note. Table extracted from De Jonckere and Rosseel (2022). Standard lower and upper bounds when fixing the variance of the factor to unity for identification

it is implemented in most statistical software is performed using an iterative optimization² algorithm where an initial solution is improved through a series of iterations. The goal of the optimization algorithm is to find those values for θ that minimize the discrepancy function in (1).

Starting values and optimization

It is important to acknowledge that numerical optimization methods can be sensitive to the initial parameter values chosen at the start of the optimization process. This stands in contrast to closed-form solutions, which possess well-defined analytical expressions that can be derived algebraically and do not exhibit sensitivity to the initial parameter values. How the starting values for the optimization process are chosen typically depends on the statistical software that is used.

Random starting values using bounds

In this section, we outline a procedure to generate random starting values for all the free parameters in a model. The only class of parameters for which we will not generate random starting values are the regression coefficients. Their starting values will always be set to zero. There are several reasons for this. First, having good starting values for these regression coefficients is far less critical (to obtain convergence) than having good starting values for the other parameters (in particular the variances). Second, by setting the regression coefficients to zero, all latent variances are total variances, not residual variances, and this simplifies the procedure.

The procedure starts with the computation of lower and upper bounds for all the (residual) variances in the model. For this, we follow De Jonckere and Rosseel (2022) and use the upper and lower bounds as shown in Tables 1 and 2, depending on how we set the metric for the latent variables. In these tables, $\text{Var}(f)$ refers to the variance of a latent variable f , while $\text{Var}(\epsilon_p)$ refers to the residual variance of the p -th indicator for this latent variable. For more details about the

² For a concise overview of optimization and optimization methods, see Appendix A.

Table 2 Bounds for ULI

Parameter	Lower bound	Upper bound
$\text{Var}(f)$	$\psi^{(l)}$	$\text{Var}(y_1)$
λ_1	1	1
λ_p	$-\sqrt{\text{Var}(y_p)/\psi^{(l)}}$	$\sqrt{\text{Var}(y_p)/\psi^{(l)}}$
$\text{Var}(\epsilon_p)$	0	$\text{Var}(y_p)$

Note. Table extracted from De Jonckere and Rosseel (2022). Standard lower and upper bounds when fixing the first-factor loading to unity for identification. The lower bound for the factor variance $\psi^{(l)}$ is defined as $\psi^{(l)} = \text{Var}(y_1) - [1 - \text{Rel}(y_1)]\text{Var}(y_1)$, where $\text{Rel}(y_1)$ is a user-specified value for the reliability of the marker indicator y_1

rationale behind these bounds, see De Jonckere and Rosseel (2022).

The next step is to compute a positive definite covariance matrix for the latent variables. To accomplish this, we first compute random correlation values (uniform between -0.5 and $+0.5$) for all the (free) covariance elements in the matrix. Then, we compute random starting values for the latent variances. We do this by drawing values from a uniform distribution, given the pre-computed lower and upper bounds for these latent variances. These (strictly positive) values are then used to convert the correlation matrix to a covariance matrix (which will always be positive definite). Next, we compute random starting values for the residual variances of the observed indicators. Again, we draw values from a uniform distribution, given the pre-computed lower and upper bounds. If we have any covariances among the residual error terms, we first generate correlation values (uniform between -0.5 and $+0.5$), and then convert them to covariances. This again ensures that the (co)variance matrix of the error terms is positive definite. Finally, we compute random starting values for the factor loadings. We use similar formulas as in Tables 1 and 2, except that $\text{Var}(y_p)$ is replaced by the starting value of the residual variance of this indicator, and $\psi^{(l)}$ is replaced by the starting value computed for the variance of this factor.

This procedure generates (bounded) random starting values that almost always result in a model implied (co)variance matrix for the observed variables that is positive definite. In the unlikely event that the (co)variance matrix is not positive definite, optimization will halt immediately, and a new set of random starting values can be computed. The procedure has been implemented in the R package lavaan (Rosseel, 2012) in version 0.6-18 or higher.³

³ This feature can be activated by providing the optional argument `rstarts = x` in the `sem()`, `cfa()`, or `lavaan()` fitting functions, where `x` is the number of sets of random starting values that will be tried. For example, `fit <- cfa(HS.model, data = HolzingerSwineford1939, rstarts = 10, verbose = TRUE)`. If multiple runs converge, we retain the solution that resulted in the smallest value for the discrepancy function.

The impact of random starting values using bounds

To evaluate the impact of choosing different starting values for nonconverged models, we generate data from a population model as shown in Fig. 1, Figs. 2 and 6.⁴ The population values are intentionally chosen to generate a high nonconvergence rate (e.g., very low reliability of the indicators). In a first study, we look at the convergence rate, improper solutions, and bias for a simple model. The second study evaluates these random starting values for a SEM model including crossloadings. The final simulation study looks at the effect of picking random starting values for a multistate-single-trait model (MSST model). Attaining convergence is an important and desirable step, yet the question about the quality of the converged solution may arise. While the primary focus of this paper is on enhancing the convergence rate, for Studies 1 and 2, we will also examine the presence of improper solutions (e.g., negative variances; see for instance Savalei and Kolenikov 2008, Chen et al. 2001 and Bollen 1987) and the relative mean bias of $\hat{\beta}$, to assess the quality of the converged model. In contrast, study 3 will solely concentrate on the convergence rate.

Study 1

For the initial simulation study, we generate one thousand random datasets, each with a sample size of 20, following the population model depicted in Fig. 1. A structural equation model is fitted to each dataset. Using both the lavaan package (Rosseel, 2012) and the OpenMx package in R (Boker et al., 2011; Neale et al., 2016), only those cases that converge or fail to converge for both packages are extracted and used for further analysis.^{5,6} A case is labeled ‘converged’ only when it satisfies two conditions: firstly, the convergence status is positive and secondly, all standard errors could be computed. The latter condition is an indirect check if $H(\theta)$ is positive definite. For cases where convergence is not achieved, we generate a maximum of 150 sets of random starting values for each case, and count the number of sets required to alter the convergence status of that particular nonconverged case.⁷

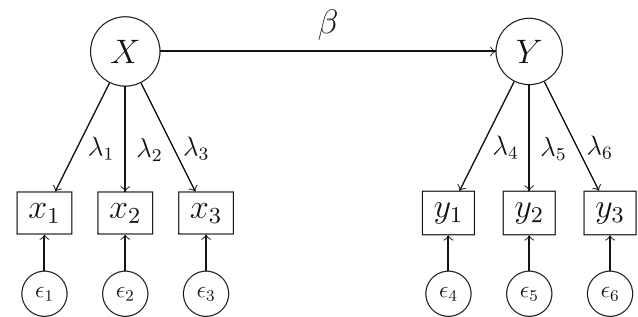


Fig. 1 Data generating model for Study 1. *Note.* A simple model where the latent variable Y is regressed on the latent variable X . Each latent variable is measured by three observed indicators. The population values are as follows: $\lambda_1 = \lambda_4 = 1$, $\lambda_2 = 0.8$, $\lambda_3 = 0.6$, $\lambda_5 = 0.7$, $\lambda_6 = 0.9$. The latent (residual) variances are set to 1.0 and the value for β equals 0.1

For the model in Fig. 1, we obtained a total of 363 nonconverged cases. The data presented in the top panel of Fig. 3 indicates that out of the 366 nonconverged cases, the majority ($N = 318$) is able to achieve convergence after 150 or fewer random sets. The top panel of Fig. 4 indicates that already 88 cases converge for one random set of starting values. Within nine sets, 50% of the 366 cases converge. These findings suggest that utilizing the bounded estimation approach to generate random sets of starting values is an efficient strategy, leading to an approximate 87%⁸ convergence rate for our initial model (Fig. 1).

Results show that 276 out of the 318 converged cases have at least one negative variance (87%; see Table 3). The maximum number of observed negative variances for a specific case is two ($N = 77$) and 42 cases have no negative variances. The mean bias for $N = 20$ and a population value for $\beta = 0.1$ is depicted in Table 4. Bias is compared between (i) nonconverged cases refitted using bounded estimation and default starting values (MLB), (ii) cases that converged in the first run (i.e., that did not need any random starting values; ML_c) and (iii) cases where bounds are constructed to form the lower and upper bound to the uniform distributions from which random starting values are drawn (without bounded estimation; RandomStart). Bias is calculated for all cases that converged within 150 or fewer random starting value sets. As can be verified in Table 4, bias for these new starting values is lower than for first run convergence using ML. A boxplot of the estimated regression coefficients can be found in Fig. 5, with a median value of -0.0001 for $\hat{\beta}$. However, the value of zero (the starting value for β) is very close to the true value for β in the model, i.e., $\beta = 0.1$. When considering a model where the true value equals 0.7, bias increases.

⁴ The code used for this paper is available for reference at <https://osf.io/8tguj/>.

⁵ By employing this approach, we aim to mitigate the influence of optimizer-specific characteristics that could potentially introduce variations in the convergence status.

⁶ For illustration, in Table 5, Appendix the reader may find results with respect to the convergence status for both lavaan and OpenMx for 10,000 replications

⁷ It takes, respectively, 5.61 s, 18.02 s, 19.18 s, 45.55 s, and 1.19 min to generate 15, 30, 50, 100, and 150 random starting value sets and fit a model as in Fig. 1

⁸ Relaxing the requirement for the starting values to yield a positive-definite matrix results in more extreme values; however, it also increases the convergence rate to 99%.

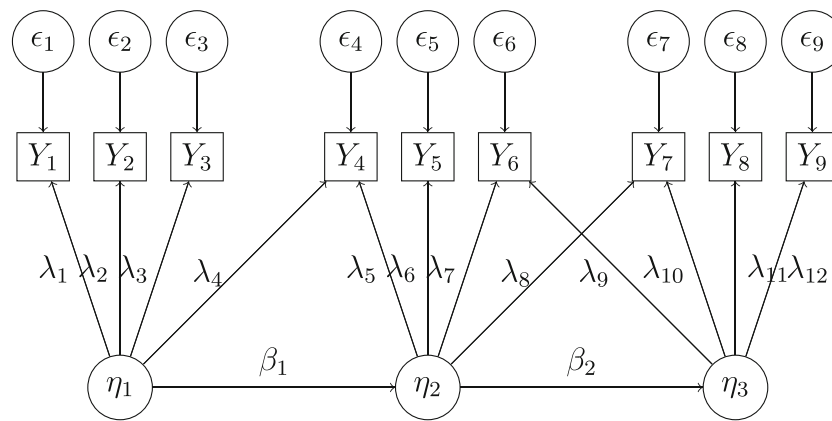


Fig. 2 Data generating model for Study 2. *Note.* The population model used for Study 2. The population values for β_1 and β_2 are 0.1 and 0.2, respectively. For the factor loadings, the population values are fixed at 1.0, except for the crossloadings (λ_4 , λ_8 , λ_9) which have a population

value of 0.3. The latent (residual) variances are set to 1.0. The residual variances of the indicators are 0.51 for the y_1 , y_2 , y_3 , y_5 , y_8 and y_9 indicators, and 0.2895 for the y_4 , y_6 and y_7 indicators

Study 2

In the second study, we consider a model containing several crossloadings (see Chen et al., 2001, p 477). As was the case for the first simulation study, we generate one thousand random datasets each with a sample size of 20, following the population model depicted in Fig. 2. Again, a structural equation model is fitted to each dataset.

In the scenario involving a crossloadings model with a total of 341 nonconverged cases, the mid panel of Fig. 3 demonstrates that employing an alternative set of starting values enables successful convergence for 25% of these cases with a single random starting value set. Roughly around 4% of nonconverged cases for this specific model is not able to converge after 150 random sets of starting values. Bias results for this simulation study show highest bias for converged solutions using random starting values, and lowest bias for converged cases using bounded estimation, i.e., the use of bounds during the model estimation phase (see the left panel of Table 4 and the right boxplot in Fig. 5). The bias for the random starting values appears extremely high for the crossloadings model. Re-calculating the mean bias for cases that only required one set results in substantially lower values, i.e., 0.08 for $\hat{\beta}_1$ and 0.53 for $\hat{\beta}_2$. The median value for $\hat{\beta}_1$ (depicted in Fig. 5) equals 0.06.⁹ For all datasets that converged within 150 random starting value sets ($N = 326$), around 90% contain at least one negative variance (see Table 3). The maximum number of observed negative variances for a specific case is five (0.9% of the converged cases).

⁹ Increasing the number of replications to 5000 (with 1701 nonconverged cases) results in a lower mean bias compared to our initial simulation study of 1000 replications (with 341 nonconverged cases). The median bias remains the same.

Study 3

We further explore the effect of bounded random starting values for a MSST model (see Fig. 6). The variance decomposition for the indicators in a MSST model is as follows:

$$\text{Var}(y_{is}) = \lambda_{s_1}^2 \lambda_{T_{s1}}^2 \text{Var}(\xi_1) + \lambda_{s_s}^2 \text{Var}(\zeta_s) + \text{Var}(\epsilon_{is}), \quad (2)$$

where $\text{Var}(\xi_1)$ the variance of the trait variable, $\lambda_{T_{s1}}$ the trait loading, $\text{Var}(\zeta_s)$ the variance of the disturbance for the state variable, and λ_{s_s} the state loading. Bounds are constructed similarly as before, but the starting values for the second-order factor loadings are set to one (i.e., $\lambda_{T_{11}}^2 = \lambda_{T_{21}}^2 = \lambda_{T_{31}}^2 = 1$). The lower and upper bounds for the variance of the higher-order trait variable are set to zero and the highest observed variance, respectively.¹⁰ A thousand random datasets, each of sample size 10, are generated. After fitting a SEM to each dataset, we find 385 nonconverged cases. For those 385 cases, random starting value sets are generated until convergence is reached (with a maximum allowed total of 150 sets). Results show that about 57% of those 385 cases converge within 150 or fewer random starting value sets ($N = 221$). All of these cases have at least one negative variance. The maximum number of observed negative variances for a specific case is four ($N = 24$).

Discussion

Nonconvergence is a well-known problem in the field of small sample structural equation modeling. While the small

¹⁰ As with the (structural) regression coefficient(s), no bounds are imposed on the second-order factor loadings. When constructing the random starting values, both the (structural) regression coefficient and these second-order parameters are ignored and set to 0 or 1, respectively.

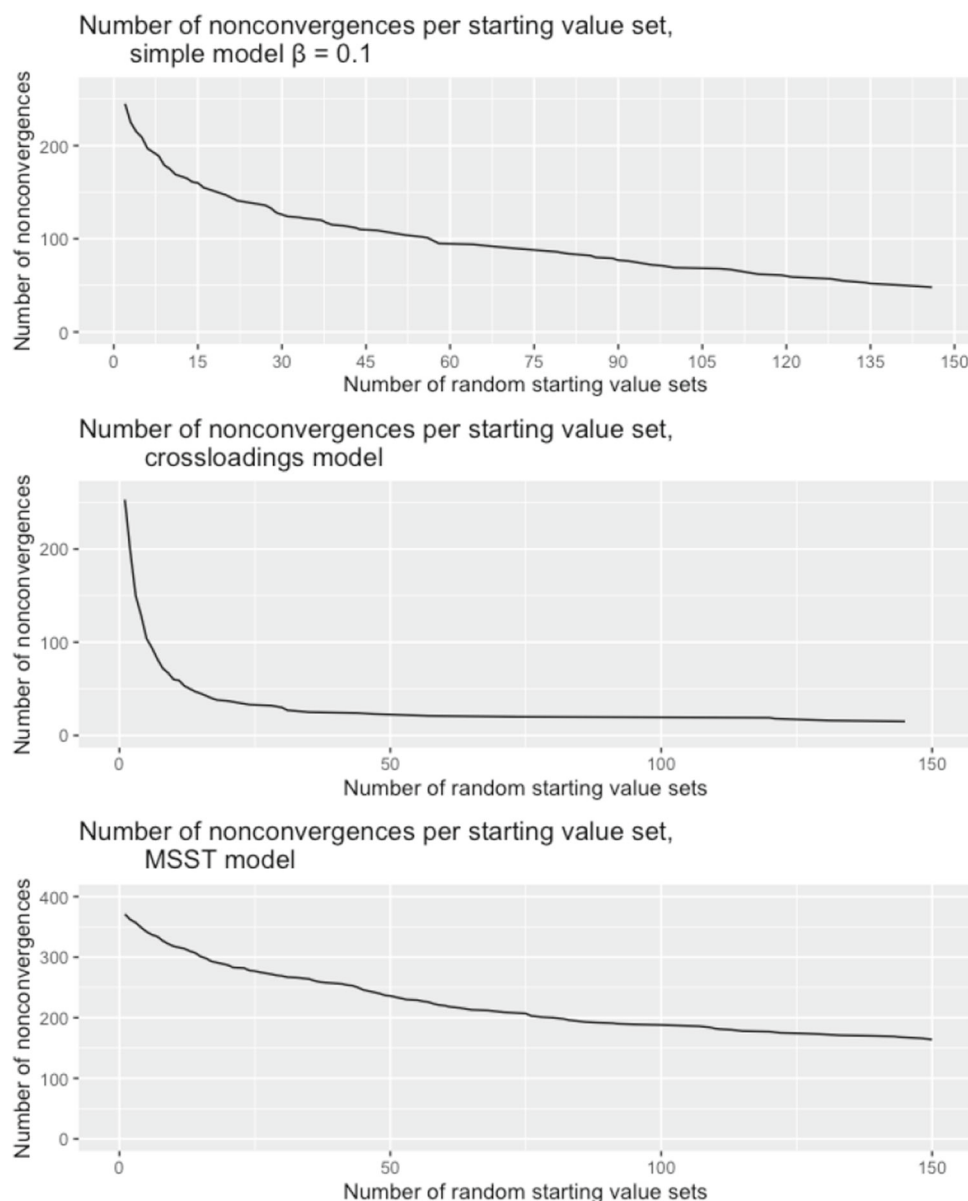


Fig. 3 Nonconverged Cases. *Note.* Number of nonconvergences per new random starting value set for a model as in Fig. 1 (top panel), Figures 2 (mid panel), and 6 (bottom panel). The total number of non-

convergences for the first model is 366, 341 for the second model, and 385 for the third model. On the Y-axis, the number of nonconverged cases for a given number of random sets (X-axis) is found

Table 3 Proportion negative variances

β	Study 1			β	Study 2		
	ML_c	RandomStart	MLB		ML_c	RandomStart	MLB
0.1	0.80 (2)	0.87 (2)	0 (0)	0.1	0.87 (5)	0.90 (5)	0 (0)
0.7	0.75 (2)	0.84 (2)	0 (0)	0.2	0.87 (5)	0.90 (5)	0 (0)

Note. Proportion negative variances in the first two studies. Results are presented for 1) first-run convergence with standard maximum likelihood (ML) and default starting values in lavaan (ML_c), 2) cases where convergence was not achieved with standard ML and default starting values, and new random starting values were constructed within upper and lower bounds (RandomStart), and 3) bias in cases where convergence was not achieved with standard ML and default starting values, and bounds were imposed on the (free) model parameters during model estimation (see De Jonckere and Rosseel (2022); MLB). The maximum number of observed negative variances per case can be found within parentheses. The number of replications on which the proportion was calculated can be found between parentheses in Table 4

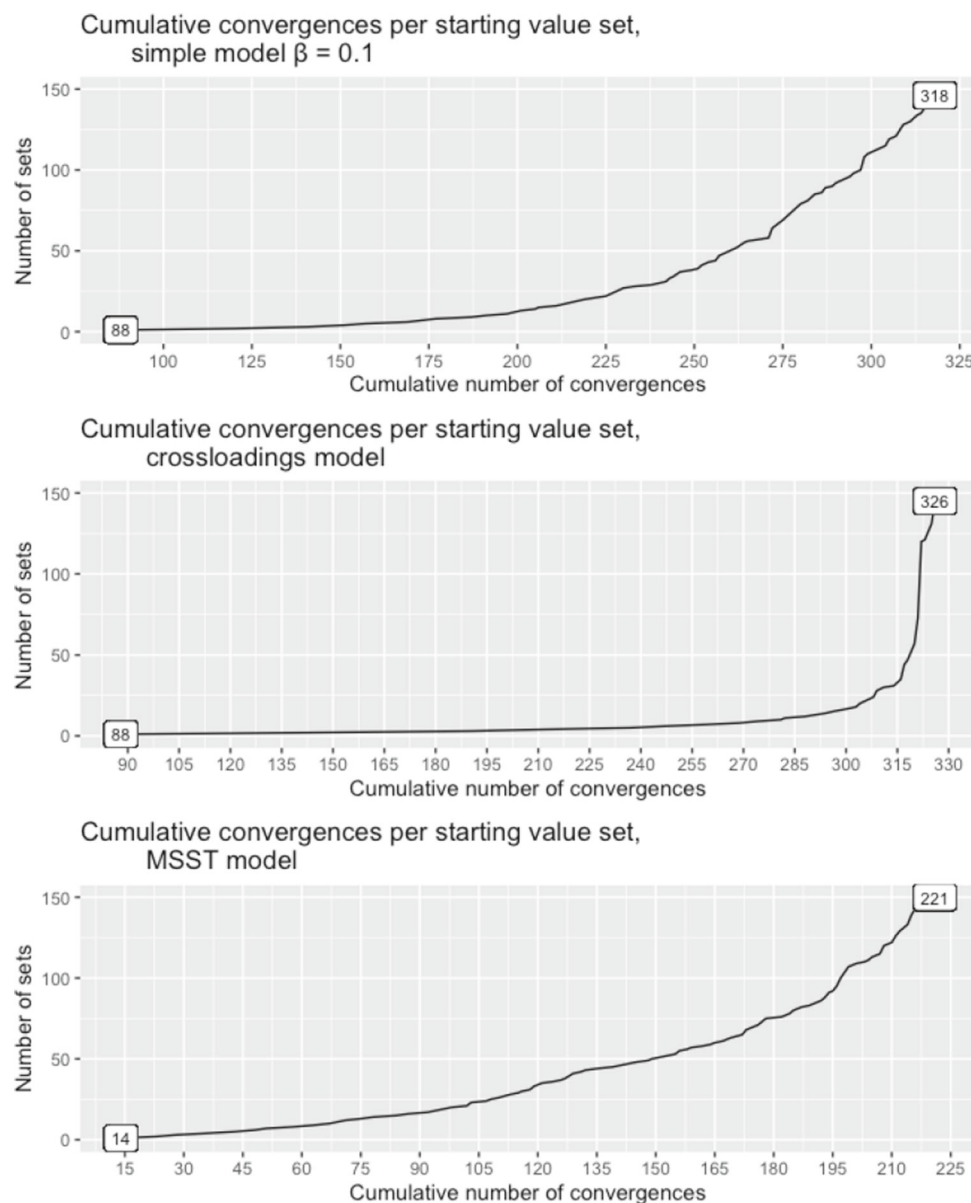


Fig. 4 Converged cases. *Note.* The cumulative number of converged cases for one to 150 random starting value sets. Results are shown for a model as in Fig. 1 (*top panel*), Figs. 2 (*mid panel*), and 6 (*bottom panel*)

sample size is often attributed as the cause, other underlying factors may also contribute. In SEM, model estimation involves an iterative procedure using optimization techniques, which necessitates the selection of appropriate starting values. The sensitivity of these techniques to the choice of starting values creates a possibility for nonconvergence when “poor” starting values are used. Although there exists a vast body of literature on optimization techniques and various approaches to determine starting values, the utilization of this knowledge in regular (non-mixture) SEMs remains limited. This paper aimed at bridging this gap by examining a specific method for determining starting values in regular SEMs: generating random starting values. These

random starting values are drawn from a uniform distribution within specified lower and upper bounds as proposed by De Jonckere and Rosseel (2022).

Our simulation studies have demonstrated that generating random starting values within data-driven bounds effectively reduces the occurrence of nonconverged solutions. Convergence was achieved in more than 80% of cases for both the first and second studies, indicating that, for most datasets, at least one set of starting values could be identified that influenced the convergence status. A third study demonstrated that even for a more complex model, such as a multistate-single-trait model, the rate of nonconvergence decreased significantly, albeit to a lesser extent. While achieving model

Table 4 Bias results

β	Study 1			β	Study 2		
	ML _c	RandomStart	MLB		ML _c	RandomStart	MLB
0.1	(307)	(318)	(366)	0.1	(659)	(326)	(341)
	-0.51	0.15	-0.07		0.18	2.53	0.03
	(346)	(286)	(327)		(659)	(326)	(341)
0.7	-0.09	-0.10	-0.02	0.2	-0.013	-0.04	-0.02
	-0.31	-0.47	-0.52		0.09	-7.16	0.06
	-0.39	-0.68	-0.50		-0.11	-0.14	-0.06

Note. Bias in the first two studies. Results are presented for 1) first-run convergence with standard maximum likelihood (ML) and default starting values in lavaan (ML_c), 2) cases where convergence was not achieved with standard ML and default starting values, and new random starting values were constructed within upper and lower bounds (RandomStart), and 3) bias in cases where convergence was not achieved with standard ML and default starting values, and bounds were imposed on the (free) model parameters during model estimation (see De Jonckere and Rosseel (2022); MLB). The first row of each β value represents the mean bias, while the second row shows the median bias. The total number of cases on which the bias was calculated can be found between parentheses

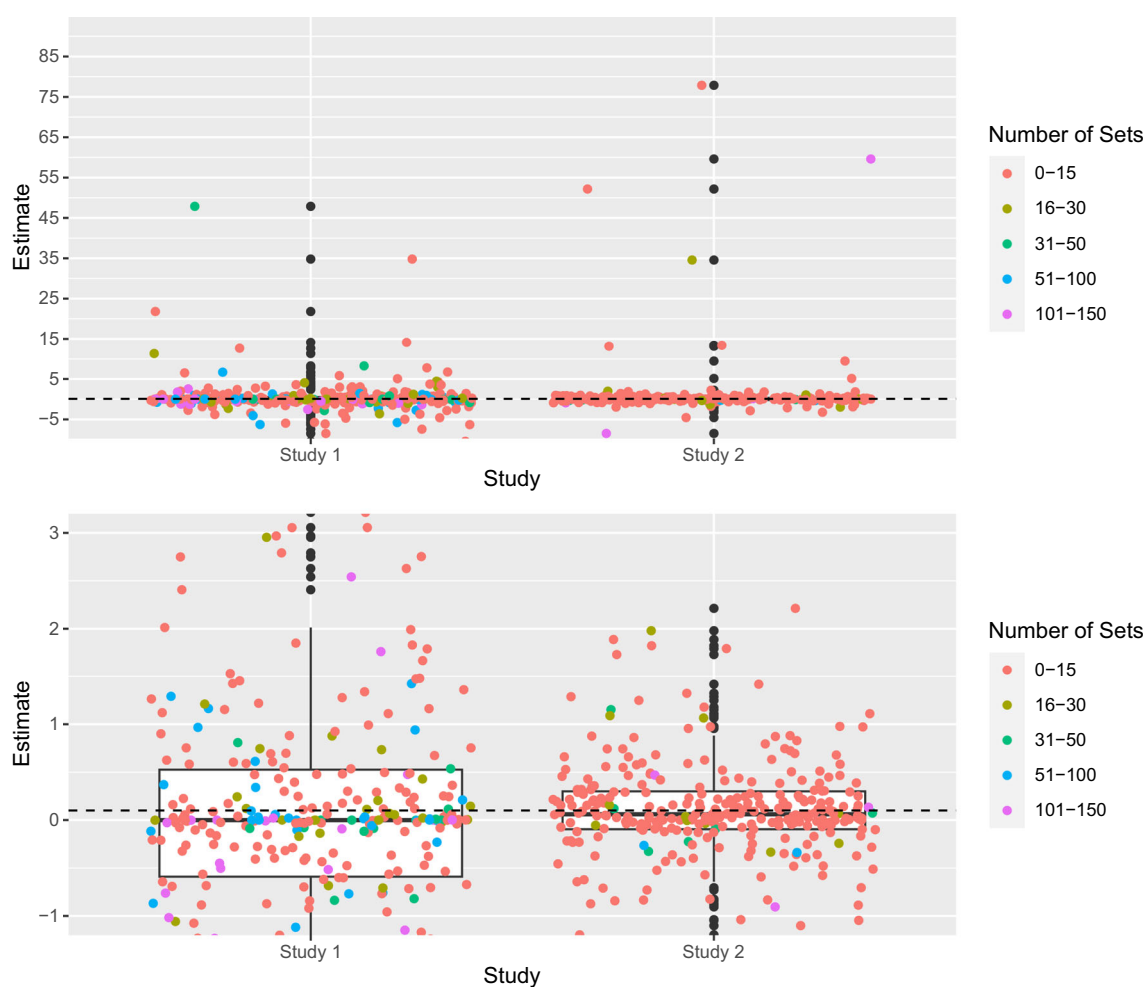


Fig. 5 Boxplot estimated structural regression coefficient. *Note.* Boxplot displaying the estimated (structural) regression coefficient for cases where bounded random starting values were used (Studies 1 and 2; we zoom in on the boxplot in the bottom panel). The black, dashed line reflects the true value for β , i.e., $\beta = 0.1$. For the second study, only the

estimates for the first (structural) regression coefficient are presented (as the true value for this estimate equals 0.1). Raw data are included, with different colors representing the number of sets required to obtain the point estimate

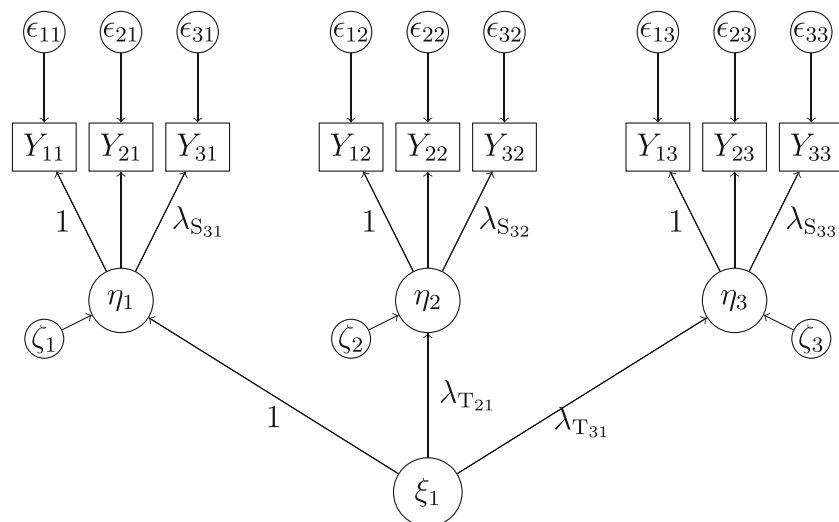


Fig. 6 Data generating model for Study 3. *Note.* Simplified representation of a multistate-single-trait (MSST) as a second-order CFA model (or a single-level SEM) for $t = 3$ timepoints, where ζ_i reflects the state variable and ξ_i the trait variable. The factor loadings are set to

$\lambda_{S_{1t}} = \lambda_{T1} = 1$, $\lambda_{S_{2t}} = \lambda_{T2} = 0.5$, and $\lambda_{S_{3t}} = \lambda_{T3} = 1.3$. The measurement errors are set to $\epsilon_{1t} = 0.6$, $\epsilon_{2t} = 0.25$, and $\epsilon_{3t} = 0.7$. The remaining true values can be consulted in the OSF repository

convergence is advantageous, given the constraint of small sample sizes, it is not a surprise that the obtained solution may not be ideal. However, it was not our objective (in this paper) to strive for an admissible solution. Therefore, preventing improper solutions or preventing extreme parameter values was not our primary goal. Getting a solution, however, even if it is a suboptimal one, is better than getting no solution at all. Even if the solution is not “ideal”, due to the presence of improper solutions or extreme parameter values, the researcher is at least able to examine the solution. In the presence of such improper solutions or extreme parameter values, the researcher may decide that the model has a poor fit. Subsequently, the researcher might opt for a technique to ‘improve’ the (quality of the) solution. For instance, by combining random starting values with bounded estimation (De Jonckere and Rosseel, 2022).

Overall, the implementation of bounded random starting values appears to offer a valuable addition to the default starting values that are currently used in software. While the default starting values can only be employed once for a given dataset, bounded random starting values can be recalculated multiple times until convergence is achieved. Moreover, it is possible that several distinct random starting value sets could lead to convergence for a single dataset. Among these “successful” starting value sets, there might exist an optimal set that produces admissible solutions with minimal bias. However, determining such an optimal set appears to be considerably complex, necessitating further simulation studies. Future research could explore this intriguing aspect. In addition, for certain datasets, bounded random starting values (and default starting values in lavaan) may fail to induce a change in the convergence status. This raises questions

about the specific data characteristics contributing to this phenomenon. We are currently exploring this further.

Finally, note that the issue of nonconvergence (and starting values) for SEM analysis disappears when noniterative estimation procedures are employed. A recent simulation study underscores the potential efficacy of these techniques (Dhaene and Dhaene, 2023).

Open Practices Statement

All code (for analysis and data generation) is accessible through the Open Science Framework using the following link: <https://osf.io/8tguj/>.

Appendix. Optimization

In general, an optimization problem is defined as

$$\underset{\theta}{\text{minimize}} F(\theta), \quad (3)$$

where F is the objective function and θ the parameter vector, subject to $\theta \in \Theta$ where Θ represents the parameter space.¹¹ Different optimization algorithms apply different techniques to explore the parameter space and to find the best possible set of θ values. Roughly speaking, these optimization algorithms can be divided into two categories: gradient-based algorithms and gradient-free algorithms.

¹¹ Note that optimization could also include maximizing an objective function, by simply switching the sign of the objective function.

Table 5 Classification table packages

lavaan	OpenMx	
	0	1
0	3701	2551
1	773	2975

Note. Convergence status for lavaan and OpenMx for rep = 10,000. A convergence status of zero signals the model did not converge

Gradient-based algorithms

Gradient-based or derivative-based algorithms utilize derivatives of the objective function while searching for the optimal solution. These algorithms differ in their utilization of either only the first derivative (gradient, $g(\theta)$), or both the first and second derivative (Hessian, $H(\theta)$). Gradient-based algorithms begin by selecting an initial point (i.e., the starting values) and evaluating the objective function at that specific point. The derivative is then computed to determine the new search direction. In the next step, the distance along the determined direction for the selection of the next point needs to be defined. This distance parameter is commonly known as the step length or the learning rate (α). The next point is then determined as a function of the current point, the search direction, and the step size. Iteratively, these initial two steps are repeated until certain convergence criteria are satisfied. Line search offers a means of selecting an optimal step size. Different types of line search exist: golden section search, bisection method, quadratic fit search, etcetera (Kochenderfer and Wheeler, 2019).

Gradient-based methods in optimization typically employ four primary convergence checks or termination conditions (Kochenderfer and Wheeler, 2019). The first condition involves terminating the process if the maximum allowable number of iterations is exceeded. The second condition evaluates whether the change in the function value for successive iterations is smaller than a predetermined threshold, i.e.:

$$f(\theta^{(k)}) - f(\theta^{(k+1)}) < \epsilon. \quad (4)$$

If the change is indeed smaller than a specified threshold (ϵ), the process is terminated. The third check also incorporates the change in the function value as in (4) but additionally utilizes the step factor in relation to the current function value, i.e.: $f(\theta^{(k)}) - f(\theta^{(k+1)}) < \epsilon |f(\theta^{(k)})|$. The last check examines whether $g(\theta^{(k+1)}) < \epsilon$. If all gradient values are lower than a certain threshold, the process is terminated. A zero gradient vector signifies the presence of a (local) minimum.

Examples of first-order methods (methods that use gradient information only) are gradient-descent and conjugate gradient (Kochenderfer and Wheeler, 2019). The Hessian matrix provides information about the curvature of a function at a specific point. This information plays a crucial role

in determining the step length for a specific parameter. By understanding the magnitude of a small or large step in relation to a specific parameter (by knowing its scale), we can conduct a more effective search for the minimum. Second-order methods differ in whether the Hessian itself (Newton's method) or an approximation of the (inverse) Hessian is used (quasi-Newton method). Calculating the Hessian at each iteration can be computationally intense, therefore, "quasi-Newton" methods (for example, BFGS, Kochenderfer and Wheeler 2019) are often preferred in practice.

Gradient-free algorithms

Gradient-free algorithms provide an alternative approach to optimization that does not rely on derivatives for determining the search direction and step size. This characteristic makes them suitable for optimizing discrete or discontinuous problems where derivative-based methods may not be applicable. Various gradient-free algorithms have been developed, including genetic algorithms, simulated annealing, and the Nelder–Mead downhill Simplex method, among others (Kochenderfer and Wheeler, 2019). These algorithms differ in their strategies for bracketing, expanding, or shrinking the search region in pursuit of an optimal solution. By avoiding the need for derivative computations, gradient-free algorithms offer flexibility in tackling a wide range of optimization problems. Nevertheless, it is worth noting that these algorithms are generally perceived as being slower compared to their gradient-based alternatives.

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Availability of data and materials / Code availability All code (for analysis and data generation) is accessible through the Open Science Framework using the following link: <https://osf.io/8tguj/>.

Declarations

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